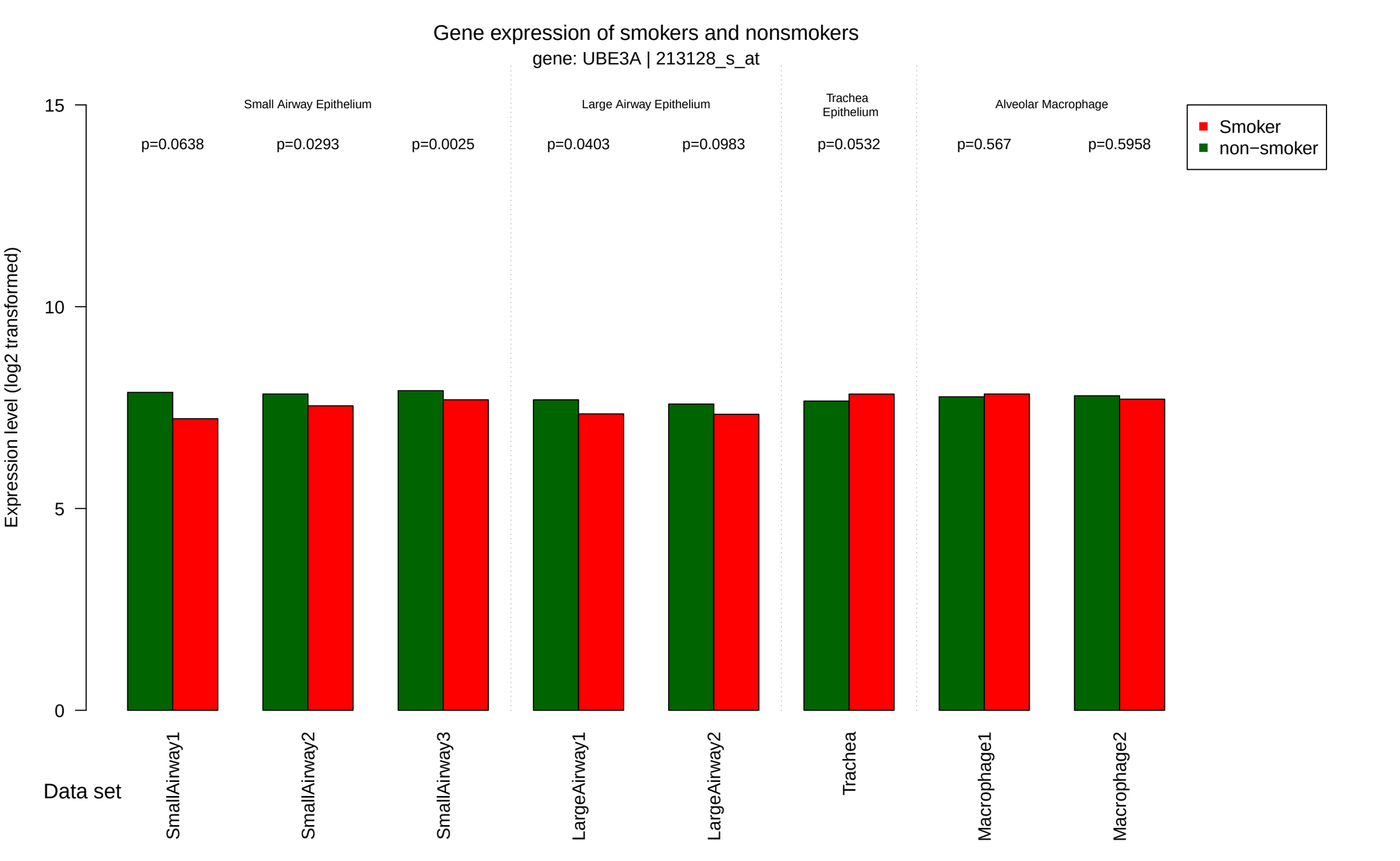
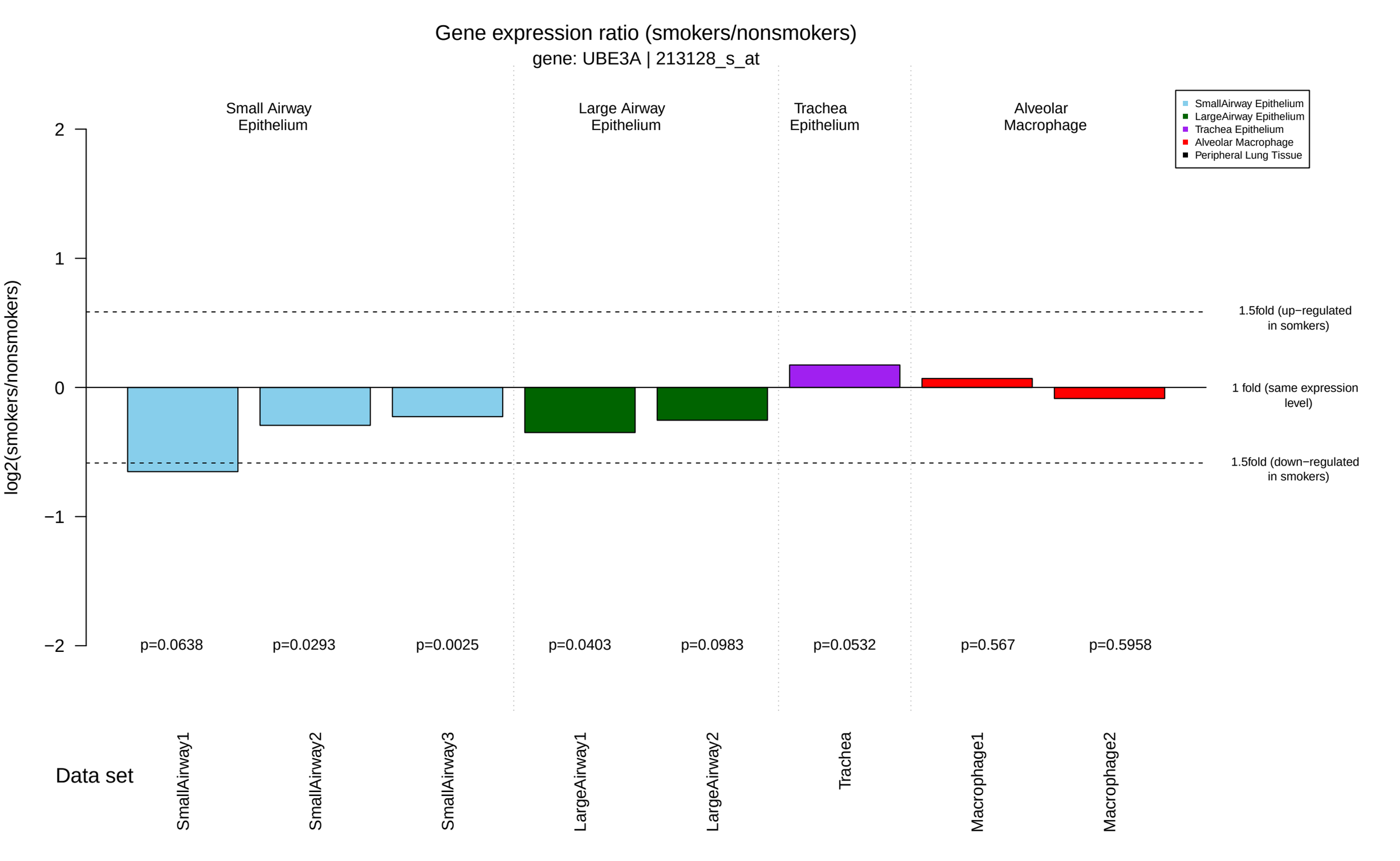
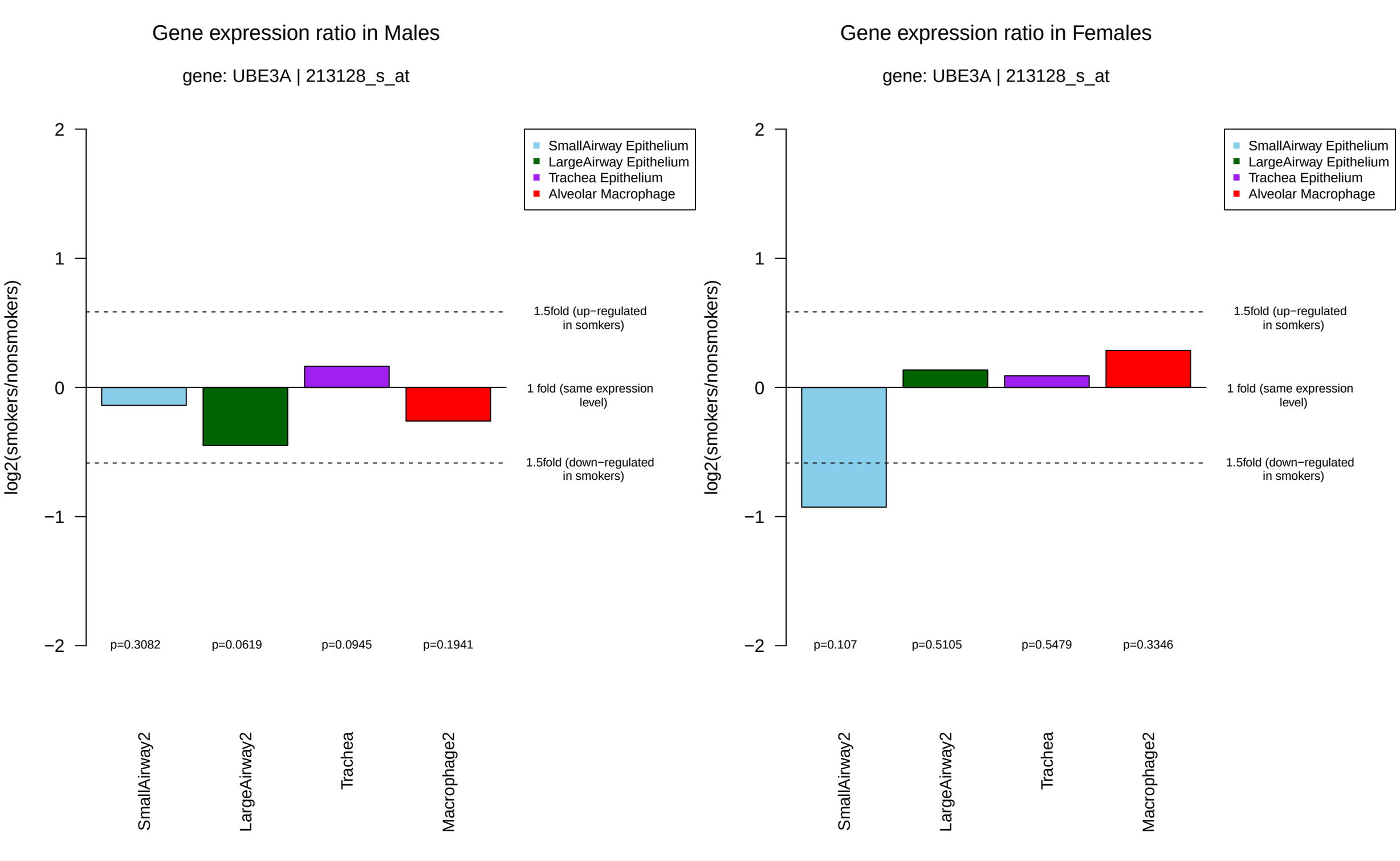
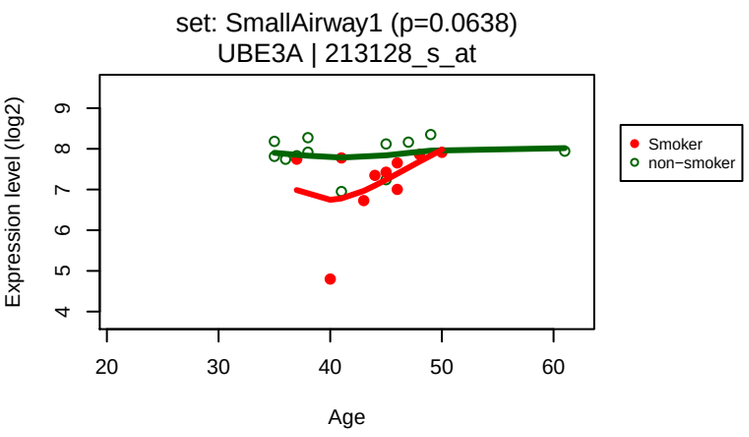
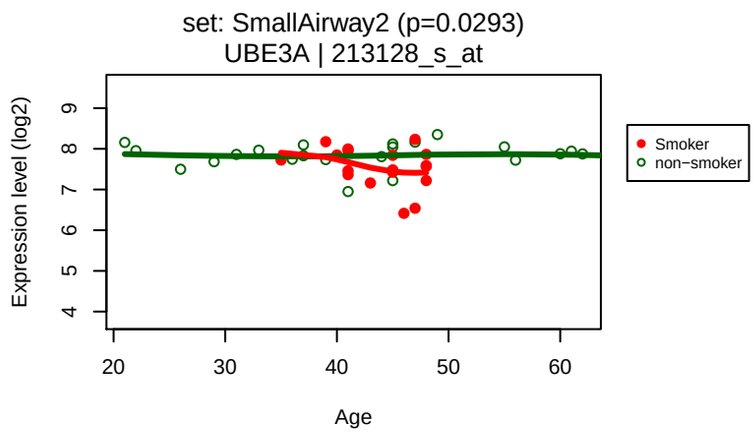
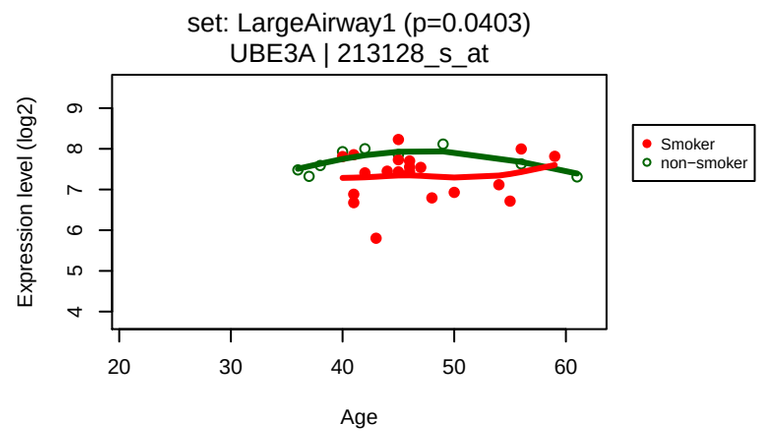

Gene expression of smokers and nonsmokers
gene: UBE3A | 213128_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0638
p=0.0293
p=0.0025
p=0.0403
p=0.0983
p=0.0532
p=0.567
p=0.5958
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

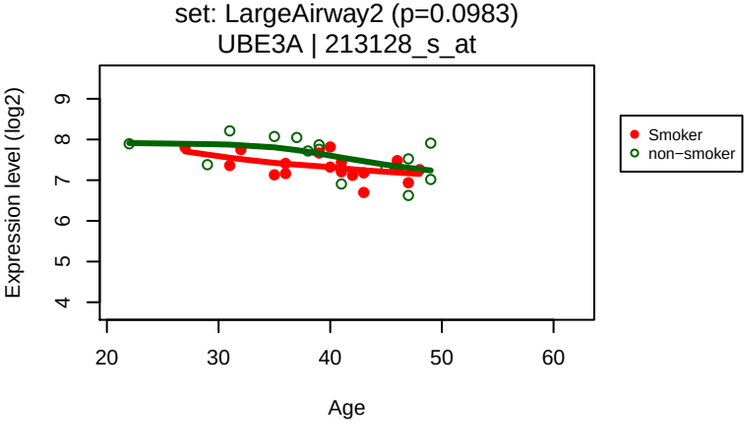

Gene expression ratio (smokers/nonsmokers)
gene: UBE3A | 213128_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0638
p=0.0293
p=0.0025
p=0.0403
p=0.0983
p=0.0532
p=0.567
p=0.5958
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

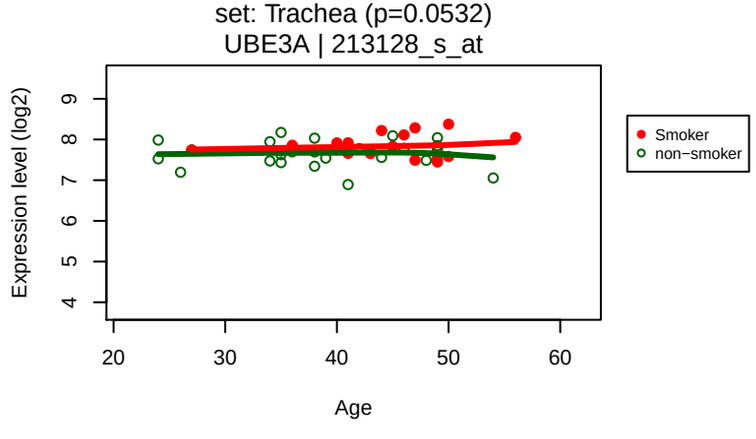

Gene expression ratio in Males
gene: UBE3A | 213128_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3082
p=0.0619
p=0.0945
p=0.1941
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: UBE3A | 213128_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.107
p=0.5105
p=0.5479
p=0.3346
SmallAirway2
LargeAirway2
Trachea
Macrophage2

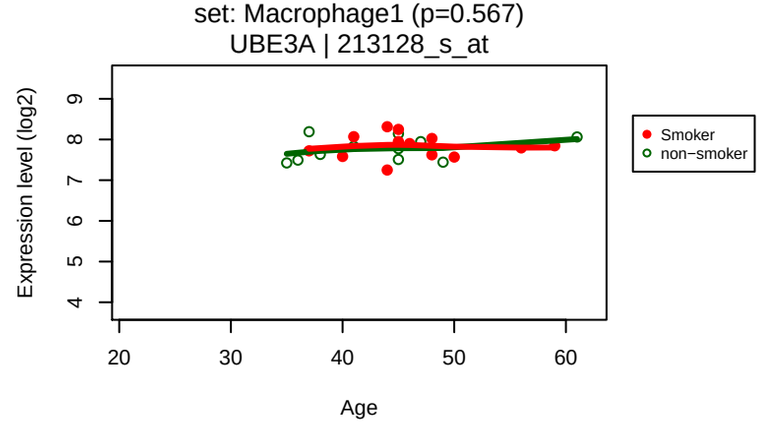

set: SmallAirway1 (p=0.0638)
UBE3A | 213128_s_at
Expression level (log2)
Age
Smoker
non−smoker

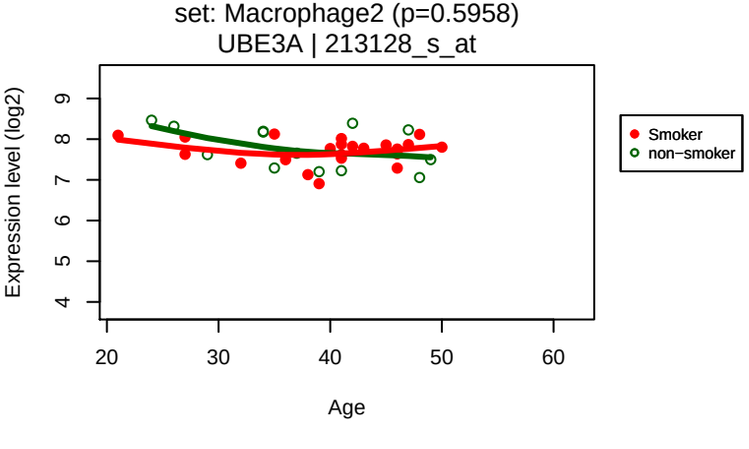

set: SmallAirway2 (p=0.0293)
UBE3A | 213128_s_at
Expression level (log2)
Age
Smoker
non−smoker

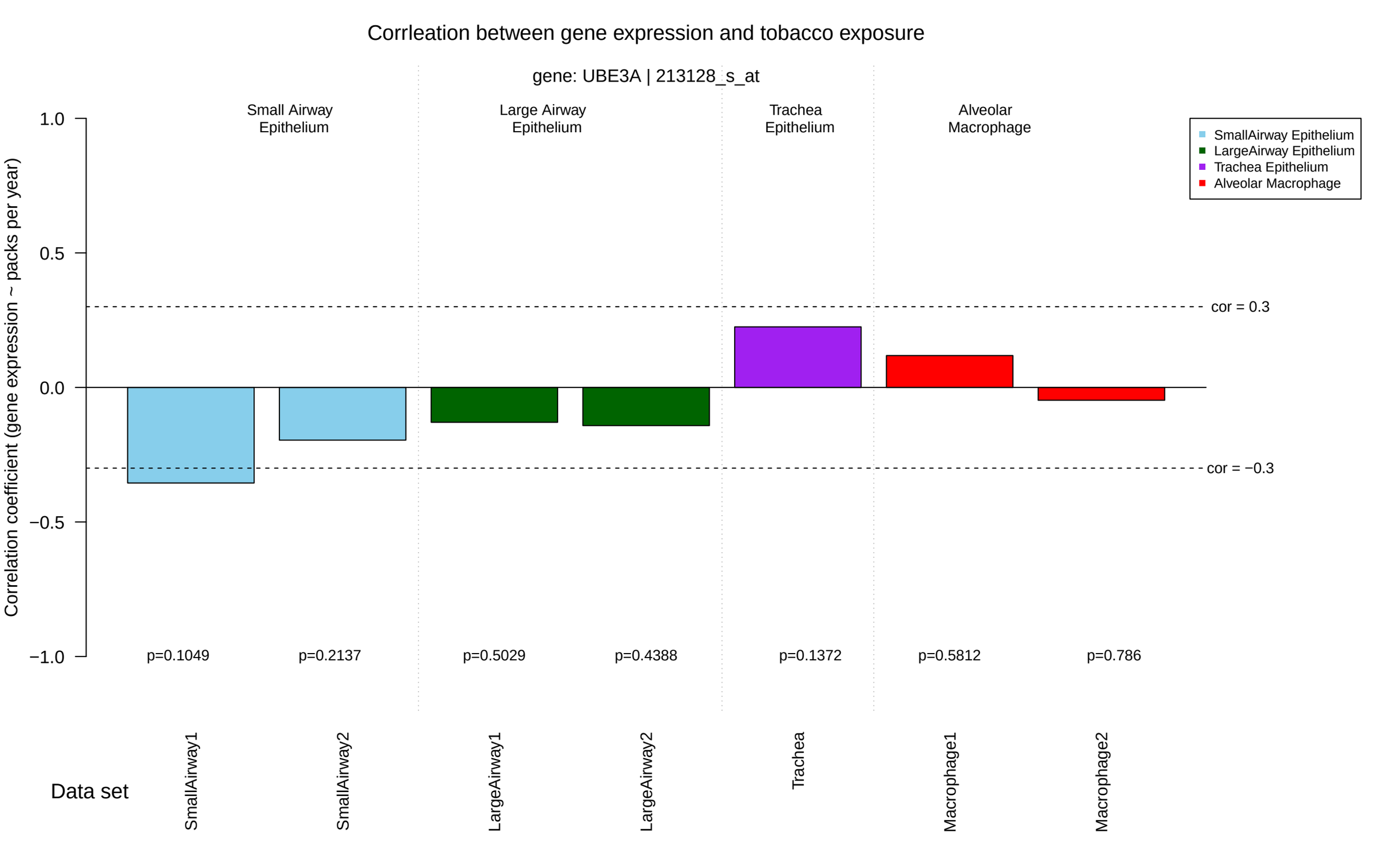

set: LargeAirway1 (p=0.0403)
UBE3A | 213128_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0983)
UBE3A | 213128_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0532)
UBE3A | 213128_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.567)
UBE3A | 213128_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5958)
UBE3A | 213128_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: UBE3A | 213128_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1049
p=0.2137
p=0.5029
p=0.4388
p=0.1372
p=0.5812
p=0.786
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2